Accompanied by his shining loved one, Beatrice, Dante ascends these spheres in the course of a fantastic, visionary journey up to the outermost sphere. When he reaches it, he contemplates the universe below him with its rotating heavens and the Earth, very far down, at its centre. But then he looks even higher – and what does he see? He sees a point of light surrounded by immense spheres of angels, that is to say, by another immense ball, which, in his words 'surrounds and is at the same time surrounded by' the sphere of our universe! Here are Dante's verses from Canto XXVII of the *Paradiso*. Q*uesta altre parte dell'Universo d'un cerchio lui comprende si come questo li altri*: 'This other part of the universe surrounds the first in a circle like the first surrounds the others.' And in the next canto, still on the last 'circle', *parendo inchiuso da quel ch'elli inchiude*: 'appearing to be to be enclosed by those that it encloses'. The point of light and the sphere of angels are surrounding the universe, and at the same time they are *surrounded by* the universe! It is an exact description of a 3-sphere!

The usual representations of Dante's universe common in Italian schoolbooks (such as figure 3.13) place the angelic spheres separate from the celestial ones. But Dante writes that the two balls 'surround and are surrounded by' each other. Dante has a clear geometrical intuition of a 3-sphere.[fn18]

The first to notice that the *Paradiso* describes the universe as a 3-sphere was an American mathematician, Mark Peterson, in 1979. In general, scholars of Dante are not very familiar with 3-spheres. Today, every physicist and mathematician could easily recognize the 3-sphere in Dante's description of the universe.

How is it possible that Dante had an idea that sounds so modern? I think it was possible, in the first place, due to the profound intelligence of Italy's finest poet. This intelligence is one of the reasons why the *Commedia* is so fascinating. But it is also due to the fact that Dante was writing well before Newton convinced everyone that the infinite space of the cosmos was the flat one of Euclidean geometry. Dante was free of the restraints upon our intuition we have as a result of our Newtonian schooling.

Dante's scientific culture was based principally on the teachings of his mentor and tutor, Brunetto Latini, who has left us a small, enchanting treatise, *Li tresor*, which is a sort of encyclopaedia of medieval knowledge, written in a delightful combination of old French and Italian. In *Li tresor*, Brunetto explains in detail the fact that the Earth is round. But he does so, curiously to the eyes of a modern reader, in terms of 'intrinsic' rather than 'extrinsic' geometry. That is to say, he does not write, 'the Earth is like an orange', as the Earth would look if seen from the outside, but writes instead, 'Two knights who could gallop sufficiently far in opposite directions would meet up on the other side.' And: 'If he were not impeded by the seas, a man who set out to